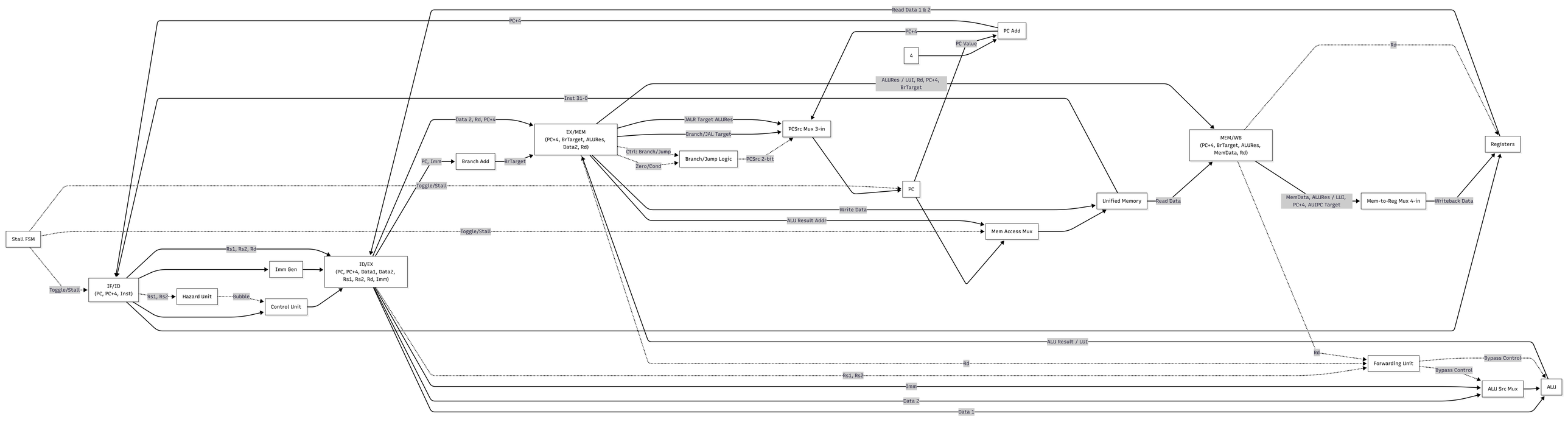
Read Data 1 & 2
PC+4
PC Add
PC+4
PC Value
4
Rd
ALURes / LUI, Rd, PC+4, BrTarget
Inst 31-0
Data 2, Rd, PC+4
EX/MEM
(PC+4, BrTarget, ALURes, Data2, Rd)
JALR Target ALURes
Branch/JAL Target
PCSrc Mux 3-in
MEM/WB
(PC+4, BrTarget, ALURes, MemData, Rd)
Registers
Ctrl: Branch/Jump
Zero/Cond
Branch/Jump Logic
PCSrc 2-bit
PC, Imm
Branch Add
BrTarget
Toggle/Stall
PC
Unified Memory
Read Data
MemData, ALURes / LUI, PC+4, AUIPC Target
Mem-to-Reg Mux 4-in
Writeback Data
Write Data
ALU Result Addr
Mem Access Mux
Stall FSM
Toggle/Stall
Rs1, Rs2, Rd
Imm Gen
ID/EX
(PC, PC+4, Data1, Data2, Rs1, Rs2, Rd, Imm)
Toggle/Stall
IF/ID
(PC, PC+4, Inst)
Rs1, Rs2
Hazard Unit
Bubble
Control Unit
ALU Result / LUI
Rd
Forwarding Unit
Bypass Control
Bypass Control
Rd
Rs1, Rs2
Imm
ALU Src Mux
ALU
Data 2
Data 1